

Learning Activity Sheet for General Science

Quarter 3

Lesson

4

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Learning Activity Sheet for General Science Grade 11
Quarter 3: Lesson 4

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Development Team

Writer: Cherry V. Abrazado PhD
Region I
Division of Alaminos City

Validator:

Language Editor:

Consultant: Crist John Pastor, PhD
Philippine Normal University

Learning Area Specialist: Dr. John Cyrus Doblada

Bureau of Learning Delivery
Bureau of Curriculum Development
Bureau of Learning Resources

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LEARNING ACTIVITY SHEET

Learning Area:	General Science	Quarter:	3
Lesson No.:	4	Date:	
Lesson Title/ Topic:	Flow Forces: Modeling Transport in Plants		
Name:		Grade & Section:	11

Objective:

Explain how water, minerals, and food are transported through the xylem and phloem, and analyze the roles of plant organs in facilitating these processes.

Part 1: Xylem Transport Model

Create a visual or physical model that shows how water moves upward through the xylem. Your model should include:

- **Transpiration pull**
- **Cohesion and adhesion**
- **Capillary action**
- **Unidirectional flow**

Guide Question:

1. What forces drive water upward through xylem, and how do they work together?

Part 2: Phloem Transport Simulation

Design a model or diagram that explains how food (sugars) moves through the phloem using the **pressure-flow hypothesis**. Your model should show:

- Source and sink dynamics
- Bidirectional flow
- Active transport and pressure gradients

Guide Questions:

1. How does the pressure-flow hypothesis explain food transport in phloem?
2. Why is phloem transport bidirectional while xylem is unidirectional?

Additional Instruction

Add color coding, arrows, and analogies to your models. For example:

- Blue arrows for water flow
- Green arrows for sugar transport
- Straws for xylem, delivery trucks for phloem

Table 1 Assessment Rubric

Criteria	Excellent (4)	Good (3)	Fair (2)	Needs Work (1)
Accuracy of Explanation	All mechanisms clearly explained	Most mechanisms explained	Some mechanisms explained	Major misconceptions present
Clarity of Model/Diagram	Highly organized and easy to interpret	Mostly clear	Some confusion	Unclear or incomplete
Depth of Reasoning	Insightful answers with justification	Reasonable answers	Basic responses	Minimal effort

Creativity and Use of Analogies	Creative and relevant analogies used	Some creative elements	Limited creativity	No analogies or visuals
Team Collaboration	All members contributed actively	Most members involved	Uneven participation	Minimal collaboration
Total Score	____/20			